

Dr. Sanam Narejo

Chairperson and Associate Professor | Department of Computer Systems Engineering

Mehran University of Engineering & Technology, Jamshoro, Pakistan

+92-3332805454 | sanam.narejo@faculty.muet.edu.pk | ORCID: 0000-0002-3537-8949

Google Scholar: scholar.google.co.id/citations?user=Qzm1c_MAAAAJ | **ResearchGate:** researchgate.net/profile/Sanam-Narejo

RESEARCH IMPACT DASHBOARD

1,268	16	22	77	53	18+
Google Scholar Citations	h-index	i10-index	Total Publications	Cumulative Impact Factor	Years of Experience

ACADEMIC QUALIFICATIONS

PhD in Electronics and Telecommunications

Politecnico di Torino, Italy | January 2014 – January 2018

Fully Funded HEC Foreign Scholarship | Thesis: Time Series Prediction and Forecasting using Deep Learning Architectures

ME in Communication Systems and Networking

Mehran University of Engineering and Technology, Pakistan | 2012 | Cumulative GPA: 3.66/4.00

Thesis: Performance Evaluation of Video Streaming Over WiMAX Network

BE in Computer Systems Engineering

Mehran University of Engineering and Technology, Pakistan | 2008 | Score: 86.5% | 2nd Position Merit Award

Final Year Project: Analysis of Quality of Service (QoS) of IEEE 802.16 in different Scheduling Services

PROFESSIONAL EXPERIENCE

Chairperson — Department of Computer Systems Engineering

Mehran University of Engineering and Technology, Jamshoro | February 2026 – Present

BS Computer Science Program Coordinator

Mehran University of Engineering and Technology, Jamshoro | November 2023 – Feb 2026

Associate Professor — Department of Computer Systems Engineering

Mehran University of Engineering and Technology, Jamshoro | January 2022 – Present

Assistant Professor — Department of Computer Systems Engineering

Mehran University of Engineering and Technology, Jamshoro | March 2018 – January 2022

Lecturer — Department of Computer Systems Engineering

Mehran University of Engineering and Technology, Jamshoro | June 2008 – March 2018

Research Fellow — Neuronica Laboratory, Department of Electronics and Telecommunications

Politecnico di Torino, Torino, Italy | March 2014 – March 2018

Research in Neural Networks, Deep Neural Networks, Feature Extraction, Signal Processing, and Time Series Prediction.

Researcher — Department of Mathematics, University of Oslo, Norway

Duration: 4 months | Data Analysis, Statistical and Machine Learning modelling for vibrational data.

Attended the International Training Workshop on Big Data and Artificial Intelligence

25–28 August 2025, Kuala Lumpur. This programme is jointly organized by **ISTIC UNESCO**.

ADMINISTRATIVE LEADERSHIP

- Chairperson, Department of Computer Systems Engineering, MUET
- BS Computer Science Program Coordinator
- NCEAC Focal Person
- PEC Program Evaluator
- OBE, CQI, Curriculum Development and Accreditation Activities
- MEMBER Boad Of Studies: BSAI, BSCYS and University of Sindh

RESEARCH SUPERVISION

1

PhD Graduated

4

Ongoing PhD Students

31

Postgraduate Students

24

UG FYP Groups

Research areas: AI, Machine Learning, Deep Learning, Healthcare AI, Computer Vision, NLP, Smart Surveillance, Robotics, IoT, Smart Agriculture, Marine Systems, Intelligent Transportation.

RESEARCH SUPERVISION — REPRESENTATIVE PROJECTS

- Human Activity Recognition using 3D Convolutional Neural Network
- Image to Text Machine Translation (Android Based Application)
- Detecting and Relieving Anxiety ON-THE-GO using Machine Learning Techniques (Funded)
- Malaria Disease Detection using a Machine Learning Algorithm
- Smart Surveillance Security System detecting weapons (Funded)
- Bitcoin Price Prediction App using Deep Learning Algorithm
- MUET FYP SMART ASSIST
- Short-Term Prediction Model for Multi-Currency Exchange Using Artificial Neural Network
- Structural Deformation Detection in Handy Skeleton
- Development of a Neural Network Based Thruster Driven Control of Ship
- Sign Language Recognition and Translation (Funded)
- Human Fall Detection using Transfer Learning
- Malicious URL Detection using Machine Learning Models
- Non-Invasive EEG Based Major Depressive Disorder Detection by Machine Learning
- An Application for Early Childhood Education through Image Recognition and Deep Learning
- SMART AI Based Boat for Marine Trash and Debris Collector
- Detection of Autism Spectrum Disorder using Machine Learning
- Autonomous Robotic Arm Grasping Objects with Reinforcement Learning
- Multi-Modal Hate Speech Recognition Through Machine Learning

PhD THESIS EXAMINER

1. DESIGN OF A DATA MINING MODEL FOR PREDICTIVE ANALYSIS OF DATA EMERGING FROM THE SECTOR OF HIGHER EDUCATION

Of Dr. Areej Fatemah

2. IMAGE PROCESSING FOR MACHINE VISION APPLICATIONS

Of Dr. Sorath Asnani

RESEARCH GRANTS & INTERNATIONAL PROJECTS

- Co-Principal Investigator — SHEC SRSP Research Grant (PKR 1.5 Million) Project on AI-based solutions for social good, funded by Sindh Higher Education Commission.
- Team Member — UNESCO-Endorsed SMART-POZZO Project (IDSSD 2026) International collaborative research project with UNESCO endorsement.

INTERNATIONAL RESEARCH EXPOSURE

- Italy — PhD at Politecnico di Torino (2014–2018)
- Spain — Advanced Statistics and Data Mining Summer Schools, UPM Madrid (2015)
- Italy — Regularization Methods for Machine Learning, DIBRIS, University of Genoa (2016)
- Italy — Italian Workshop on Neural Networks (WIRN), Salerno (2016)
- Norway — Visiting Researcher, University of Oslo (4 months)
- France, Germany & Switzerland — Academic exposure during PhD
- Malaysia — Professional Training

RESEARCH INTERESTS

- Machine Learning & Deep Learning
- Explainable AI (XAI)
- Time Series Prediction and Forecasting
- Signal Processing & EEG Analysis
- Computer Vision & Image Processing
- Natural Language Processing (NLP)
- Healthcare AI & IoT
- Generative AI
- Business Intelligence & Big Data Analytics
- Smart Agriculture & Marine Systems

TECHNICAL COMPETENCIES

Programming & Scripting:

Python, MATLAB, R, SQL, LaTeX

Deep Learning Frameworks:

TensorFlow, PyTorch, Keras

Data Science Libraries:

Scikit-learn, Pandas, NumPy, Seaborn, Matplotlib

Computer Vision & AI:

OpenCV, YOLO, Hugging Face, Transformers

Databases:

MySQL, MongoDB

Visualization & Analytics:

Power BI, Tableau, SPSS

Tools & Platforms:

Jupyter Notebook, Google Colab, Git / GitHub, Visual Paradigm, Adobe Photoshop, Inkscape

LANGUAGES

- English — Fluent
- Urdu — Native
- Italian — Proficient

PROFESSIONAL MEMBERSHIPS

- Pakistan Engineering Council (PEC)
- Association for Computing Machinery — Women (ACM-W)
- IEEE Computer Society — Faculty Advisor, MUET
- Women in STEM Committee (BWISC) — 2021–22
- SINCERE-MUET Research Group

COURSES TAUGHT

Undergraduate Level:

- Data Science & Analytics
- Artificial Intelligence
- Machine Learning
- Database Management System
- Software Engineering

Masters Level:

- Data Mining
- Knowledge Discovery and Data Mining
- Machine Learning Practical
- Deep Learning
- Research Methodologies

PhD Level:

- Advanced Machine Learning
- Machine learning Techniques
- Computational Models and Complexities

Professional / Industry Training:

- Big Data Analytics and Techniques — National Vocational & Technical Training Commission (NAVTTTC), Pakistan
- Data sciences in PITP

INVITED TALKS & PRESENTATIONS

- AI Prompting — IEEE WIE, 2026
- The World of AI and Intellectual Paralysis, 2024
- The Diversity Between Human Computer Interaction and Machine Learning, 2022
- Role of Statistics and Probability in Machine Learning and Data Science, 2021
- Artificial Intelligence in Today's World, 2018

AWARDS & DISTINCTIONS

- Best Alumni Teaching Award — February 2025
- Fully Funded HEC Foreign Scholarship for PhD — Politecnico di Torino, Italy
- Merit Certificate — 2nd Position in Bachelor of Computer Systems Engineering
- Shortlisted — AMI DLDR Summer School 2025

SUMMER SCHOOLS & WORKSHOPS

- Neural Networks and Deep Learning — Advanced Statistics and Data Mining Summer School, UPM Madrid, Spain, 2015
- Time Series Forecasting — Advanced Statistics and Data Mining Summer School, UPM Madrid, Spain, 2015
- Regularization Methods for Machine Learning — DIBRIS, University of Genoa, Italy, 2016
- The Italian Workshop on Neural Networks (WIRN), Salerno, Italy, 2016
- Shortlisted: AMI DLDR Summer School 2025

JOURNAL PUBLICATIONS (50 papers | Cumulative IF: 53)

1. Mushtaq N., Narejo S., Ali S.A., Ahmed F. and Hassan S.R. "Deep explainable CNN (DE-CNN) framework for EMG-based prosthetic control with global and local interpretability." *The Journal of Supercomputing*, 2026. [HEC: W] [IF: 3.5]
2. Akhi M., Sarang K., Narejo S., Farheen S., Pahlevanzadeh B., Dhirani L. "Towards Accessible, Inclusive, and Secure Digital Ecosystems Driven by Emerging Technologies: A Systematic Review, Conceptual Framework, and Future Research Directions." *Human Behavior and Emerging Technologies*, 2026. [HEC: W] [IF: 4.9]
3. Mustafa R., Jiskani S.M., Panhwar M.A., Narejo S. and Ahmed Z. "YOLO-Scan: A Real-Time Deep Learning Framework for Brain Tumor Detection in MRI." *Mehran University Research Journal of Engineering and Technology*, 45(2), pp.62-73, 2026. [HEC: X] [IF: 0.9]
DOI: 10.22581/0545
4. Talpur U., Memon M., Narejo S., Baloch S. and Talpur S. "Enhancing Traffic Law Enforcement through Deep Learning-Based Detection of One-Wheeler Motorcycle Stunts." *Journal of Independent Studies and Research Computing (JISR-C)* 24(1), pp.72-88, 2026. [HEC: Y]
DOI: 10.31645/JISRC.26.24.1.7
5. Memon M., Narejo S., Baloch Z., Talpur U. and Manzoor A. "An occupancy-aware IoT-based framework for real-time monitoring and optimization of energy consumption in smart homes." *Spectrum of Engineering Sciences*, 4(3), pp.3600–3613, 2026. [HEC: Y]
DOI: 10.5281/zenodo.21425169
6. Hajano S., Narejo S., Memon M., Memon M. "Deepfake Detection via Digital Forensic Artifact Analysis: A Multi-Artifact Framework." *Spectrum of Engineering Sciences*, 4(3), pp.3711–3733, 2026. [HEC: Y]
DOI: 10.5299/zenodo.2142544
7. Rai T., Kumar V., Memon M. and Narejo S. "A Sustainable Next-Generation Digital Tourism Guide for Sindh Using a Multilingual Chatbot." *International Journal of Innovations in Science & Technology*, 8(1), pp.255-267, 2026. [HEC: Y]
8. Qamar N., Narejo S., Javaid R., Sikandar N. and Hajano S. "The role of emotional intelligence in workplace leadership: How emotional intelligence influences leadership effectiveness, team collaboration, and employee well-being." *Journal of Media Horizons*, 6(4), pp.1102–1125, 2025. [HEC: Y]
DOI: 10.5281/zenodo.17309914
9. Bashir N., Narejo S., Naz B., Ismail F., Anjum M.R., Butt A., Anwar S. and Prasad R. "A machine learning framework for major depressive disorder (MDD) detection using non-invasive EEG signals." *Wireless Personal Communications*, 140(1), pp.39-61, 2025. [HEC: W] [IF: 1.9]
DOI: 10.1007/s11277-023-10445-w
10. Narejo S., Hajano S., Shaikh A., Virk J. and Shaikh M. "Efficient and Deployable Violence Detection using Transfer Learning Framework for Public Safety." *Mehran University Research Journal of Engineering and Technology*, 44(4), pp.01-16, 2025. [HEC: X] [IF: 0.9]
DOI: 10.22581/muet1982.0125
11. Ali S., Baloch Z., Narejo S. and Shaikh A.M. "A Review of Developments in Generative AI, Machine Learning, and Neuroimaging for the Diagnosis of Autism." *VFAST Transactions on Software Engineering*, 13(2), 68–82, 2025. [HEC: Y]
DOI: 10.21015/vtse.v13i2.2089
12. Mushtaq N.U.A., Narejo S., Ali S.A. and Jawaid M.M. "Explainable Machine Learning for Mental Health Detection Using NLP." *ADCAIJ: Advances in Distributed Computing and Artificial Intelligence Journal*, 14, pp.e32449-e32449, 2025. [HEC: W] [IF: 1.6]
DOI: 10.14201/adcaij.32449
13. Hajano S., Narejo S., Hajano S. and Hajano F. "A Few-Shot Learning Framework for Understanding Mobile User Behavior." *Mehran University Research Journal of Engineering and Technology*, 44(4), pp.25-31, 2025. [HEC: X] [IF: 0.9]
DOI: 10.22581/muet1982.0155
14. Jatoti S., Abro B., Narejo S., Baloch Y.A. and Asma K. "Harnessing Machine Learning for Accurate Smog Level Prediction: A Study of Air Quality in India." *VAWKUM Transactions on Computer Sciences*, 13(1), pp.106-119, 2025. [HEC: Y]
DOI: 10.21015/vtcs.v13i1.2077
15. Zaki M.A., Zai S., Amjad U., Zaki U. and Narejo S. "Compact Convolutional Neural Network Architecture for Onion Disease Classification Using Crop Images." *Advances & Applications in Discrete Mathematics*, 42(3), 2025. [HEC: Y]
DOI: 10.17654/0974165825017
16. Wagan A.A., Talpur S. and Narejo S. "Clustering uncertain overlapping symptoms of multiple diseases in clinical diagnosis." *PeerJ Computer Science*, 10, p.e2315, 2024. [HEC: W] [IF: 3.8]
DOI: 10.7717/peerj-cs.2315
17. Jawaid M.M., Narejo S., Riaz F., Reyes-Aldasoro C.C., Slabaugh G. and Brown J. "Non-calcified plaque-based coronary stenosis grading in contrast enhanced CT." *Medical Engineering & Physics*, 129, 104182, 2024. ISSN: 1873-4030. [HEC: W] [IF: 1.7]
DOI: 10.1016/j.medengphy.2024.104182
18. Ali S., Naz B., Narejo S. and Ahmed Z. "AlexNet-Based Speech Emotion Recognition Using 3D Mel-Spectrograms." *International Journal of Innovations in Science and Technology*, 6(2), 426–433, 2024. ISSN 2618-1630. [HEC: Y]
19. Azhar Z., Chaudhry H.N., Kulsoom F. and Narejo S. "Deep Learning-Based Automated Classroom Slide Extraction." *International Journal of Innovations in Science and Technology*, 6(2), 380–395, 2024. ISSN 2618-1630. [HEC: Y]
20. Memon F., Naz B., Narejo S. and Panhwar K. "An Artificial Intelligence Vision Transformer Model for Classification of Bacterial Colony." *International Journal of Innovations in Science and Technology*, 6(1), 237–248, 2024. ISSN 2618-1630. [HEC: Y]
21. Memon Z., Turab M., Narejo S. and Korejo M.T. "An ensemble of CNN architectures for early detection of Alzheimer's disease using brain MRI." *Mehran University Research Journal of Engineering & Technology*, 42(4), 140-147, 2023. [HEC: X] [IF: 0.9]
DOI: 10.22581/muet1982.2304.2767
22. Altaf Z., Unar M.A., Narejo S., Zaki M.A. and Naseer-u-Din. "Generalized Epileptic Seizure Prediction using Machine Learning Method." *International Journal of Advanced Computer Science and Applications (IJACSA)*, 14(1), 2023. ISSN 2156-5570. [HEC: W*] [IF: 1.1]
DOI: 10.14569/IJACSA.2023.0140155
23. Kulsoom F., Narejo S., Mehmood Z., Chaudhry H.N., Butt A. and Bashir A.K. "A Review of Machine Learning-based Human Activity Recognition for Diverse Applications." *Neural Computing and Applications*, 34, pp.18289–18324, 2022. [HEC: W] [IF: 5.0]
DOI: 10.1007/s00521-022-07665-9
24. Butt A., Narejo S., Anjum M.R., Yonus M.U., Memon M. and Samejo A.A. "Fall Detection Using LSTM and Transfer Learning." *Wireless Personal Communications*, 126, pp.1733–1750, 2022. ISSN 1572-834X. [HEC: W] [IF: 1.9]
DOI: 10.1007/s11277-022-09819-3
25. Unar S., Unar M.A., Memon Z.A. and Narejo S. "A Neural network based heading and position control system of a ship." *Mehran University Research Journal of Engineering & Technology*, 41(2), 172-177, 2022. [HEC: X] [IF: 0.9]
DOI: 10.22581/muet1982.2202.16
26. Ahmad N., Younus M.U., Anjum M.R., Saleem G., Gondal Z.A. and Narejo S. "Efficient JPEG Encoding Using Bernoulli Shift Map for Secure Communication." *Wireless Personal Communications*, 125, pp.3405–3424, 2022. ISSN 1572-834X. [HEC: W] [IF: 1.9]
DOI: 10.1007/s11277-022-09717-8

27. Bukhari S.S.H., Younus M.U., Hassan M.A.S., Anjum M.R. and Narejo S. "Scientific Workflow Makespan Minimization In Edge Multiple Service Providers Environment." *Wireless Personal Communications*, 125, pp.3187–3203, 2022. [HEC: W] [IF: 1.9]
DOI: 10.1007/s11277-022-09704-z
28. Bashir N., Narejo S., Naz B., Jawed M.M., Talpur S. and Aliev K. "Non-invasive EEG based Feature Extraction framework for Major Depressive Disorder analysis." *International Journal of Innovations in Science and Technology*, 4(1), pp.110-122, 2022. ISSN 2618-1630. [HEC: Y]
29. Bakhtawar, Naz B., Din N.U., Ahmed W. and Narejo S. "Soil Classification & Prediction of Crop Status with Supervised Learning Algorithm: Random Forest." *International Journal of Innovations in Science & Technology*, 4(4), pp.1011–1022, 2022. [HEC: Y]
30. Noreen A., Jawaid M., Narejo S., Talpur S. and Ayub B. "Coronavirus Detection using Deep Pre-trained Model on Chest X-ray Imaging." *Quaid-e-Awam University Research Journal of Engineering Science and Technology*, 20(1), pp.55-60, 2022. ISSN 1605-8607. [HEC: Y]
DOI: 10.52584/QRJ.2001.08
31. Narejo S., Shaikh A., Memon M.M., Mahar K., Aleem Z. and Zardari B. "Big data analytics and classification of cardiovascular disease using machine learning." *Journal of Intelligent & Fuzzy Systems*, 43(2), pp.2025-2033, 2022. [HEC: W] [IF: 1.7]
DOI: 10.3233/JIFS-219302
32. Memon M., Jawaid M.M., Narejo S. and Rathi M. "Computer-Assisted Framework for Automatic Detection of Structural Hand Deformities." *Journal of Hunan University Natural Sciences*, 48(10), 2021. ISSN 1674-2974. [HEC: X]
33. Narejo S., Jawaid M.M., Talpur S., Baloch R. and Pasero E. "Multi-step rainfall forecasting using deep learning approach." *PeerJ Computer Science*, 2021. ISSN 2376-5992. [HEC: W] [IF: 3.8]
DOI: 10.7717/peerj-cs.514
34. Zaki M.A., Narejo S., Ahsan M., Zai S., Anjum M.R. and Din N.U. "Image-based Onion Disease (Purple Blotch) Detection using Deep Convolutional Neural Network." *IJACSA*, 12(5), 2021. ISSN 2156-5570. [Scopus Q3, WOS ESCI] [HEC: W*] [IF: 1.1]
35. Narejo S., Pandey B., Vargas D.E., Rodriguez C. and Anjum M.R. "Weapon Detection Using YOLO V3 for Smart Surveillance System." *Mathematical Problems in Engineering*, vol. 2021, Article ID 9975700, 2021. [Scopus Q2] [HEC: X] [IF: 1.5]
DOI: 10.1155/2021/9975700
36. Bukhari S.S.H., Kabir A., Younus M.U., Anjum R. and Narejo S. "Novel Cost Efficient Resource Allocation Technique Based on Deadline and Budget Constraints for Edge Users." *Wireless Personal Communications*, 120, pp.269–286, 2021. ISSN 1572-834X. [HEC: W] [IF: 1.9]
DOI: 10.1007/s11277-021-08453-9
37. Zaki M.A., Narejo S., Zai S., Zaki U., Altaf Z. and Din N.U. "Detection of nCoV-19 from Hybrid Dataset of CXR Images using Deep Convolutional Neural Network." *IJACSA*, 11(12), 2020. ISSN 2156-5570. [Scopus Q3, WOS ESCI] [HEC: W*] [IF: 1.1]
DOI: 10.14569/IJACSA.2020.0111281
38. Afzal M., Jawaid M.M., Baloch R.B. and Narejo S. "Automatic Detection of Elbow Abnormalities in X-ray Imagery." *IJACSA*, 11(12), 2020. ISSN 2156-5570. [Scopus Q3, WOS ESCI] [HEC: W*] [IF: 1.1]
39. Narejo S., Talpur S., Memon M. and Rahoo A. "An automated system for traffic sign recognition using convolutional neural network." *3c Technologia*, 9(1), pp.119-135, 2020. [ESCI, WOS] [HEC: W*]
40. Narejo A., Baqai A., Sikandar N., Ali A. and Narejo S. "Physiotherapy: Design and Implementation of a Wearable Sleeve using IMU Sensor and VR to Measure Elbow Range of Motion." *IJACSA*, 11(9), 2020. ISSN 2156-5570. [Scopus Q3, WOS ESCI] [HEC: W*] [IF: 1.1]
DOI: 10.14569/IJACSA.2020.0110953
41. Narejo S., Soomro S.S., Naz B., Noreen A. and Memon N. "Development of vehicle driver drowsiness detection system using eye aspect ratio." *PalArch's Journal of Archaeology of Egypt/Egyptology*, 17(9), pp.8957-8971, 2020. [Scopus Q3] [HEC: SCOPUS]
42. Memon S., Ahmed M., Narejo S., Baig U.A., Chowdry B.S. and Anjum M.R. "Self-Driving Car Using LiDAR Sensing and Image Processing." *International Journal of Grid and Distributed Computing*, 13(2), pp.77-88, 2020. [ESCI, EBSCO] [HEC: Y]
43. Tabbassum K., Talpur S. and Narejo S. "Management of Scratchpad Memory Using Programming Techniques." *Mehran University Research Journal of Engineering and Technology*, 38(2), 2019. ISSN 0254-7821. [HEC: X] [IF: 0.9]
DOI: 10.22581/muet1982.1902.05
44. Jawaid M.M., Narejo S., Qureshi I.A. and Pirzada N. "A Review of the State-of-the-Art Methods for Plaque Detection in Cardiac CT Angiography." *International Journal of Computer Theory and Engineering*, 10(3), 2018. [HEC: X]
45. Jawaid M.M., Narejo S., Pirzada N., Baloch J., Reyes-Aldasoro C.C. and Slabaugh G. "Automated Quantification of Non-Calcified Coronary Plaques in Cardiac CT Angiographic Imagery." *IJACSA*, 9(7), 2018. [HEC recognized, IF: 1.7] [HEC: W*] [IF: 1.7]
46. Aliev K., Jawaid M.M., Narejo S., Pasero E. and Pulatov A. "Internet of Plants Application for Smart Agriculture." *IJACSA*, 9(4), 2018. [HEC: W*] [IF: 1.7]
DOI: 10.14569/IJACSA.2018.090458
47. Narejo S. and Pasero E. "Meteonowcasting using Deep Learning Architecture." *IJACSA*, 8(8), 2017. [HEC: W*] [IF: 1.7]
48. Narejo S., Pasero E. and Kulsoom F. "EEG Based Eye State Classification using Deep Belief Network and Stacked AutoEncoder." *International Journal of Electrical & Computer Engineering*, 6(6), 2016. [SCOPUS Q2] [HEC: HEC]
49. Narejo S. and Pasero E. "Time series forecasting for outdoor temperature using nonlinear autoregressive neural network models." *Journal of Theoretical and Applied Information Technology*, 94(2), 451, 2016. [SCOPUS Q4] [HEC: HEC]
50. Narejo S., Kulsoom F. and Jawaid M.M. "Performance Evaluation of Interactive Video Streaming over WiMAX Network." *International Journal of Electrical & Computer Engineering*, 7(1), 2017. [SCOPUS Q2] [HEC: HEC]

INTERNATIONAL BOOK CHAPTERS (7 chapters | All Scopus-indexed)

1. Memon M., Talpur S., Narejo S., Channa A. and Pandey J.K. "Fundamentals of image processing for surveillance." In *Computational Intelligence in Surveillance Systems Using Image Processing* (pp. 13-24). Elsevier, 2026. [Scopus-indexed]
DOI: 10.1016/B978-0-443-36408-2.00010-2
2. Memon M., Narejo S., Talpur S., Channa A., Mangi F.A. and Pandey J.K. "Explainable AI (XAI) in Air Traffic Monitoring Systems." Emerald Publishing Limited, 2026. [Scopus-indexed]
DOI: 10.1108/978-1-80592-062-520251002
3. Memon M., Talpur S., Narejo S. and Sanin C. "Development of non-GPS drone mission control for autonomous horizontal navigation." In *Computational Intelligence in Surveillance Systems Using Image Processing* (pp. 83-95). Elsevier, 2026. [Scopus-indexed]
DOI: 10.1016/B978-0-443-36408-2.00025-4
4. Khanzada A.A., Hasnain A., Narejo S., Chowdhry B.S. and Laxmi L. "Impact of Digitalization on Social Entrepreneurship." In *Two Faces of Digital Transformation*, pp. 19-29. Emerald Publishing Limited, 2023. [Scopus-indexed]
DOI: 10.1108/978-1-83753-096-020231002
5. Bashir N., Narejo S., Naz B. and Ali A. "EEG Based Major Depressive Disorder (MDD) Detection Using Machine Learning." In *Pattern Recognition and Artificial Intelligence*, MedPRAI 2021, CCIS vol. 1543. Springer, Cham, 2022. [Scopus-indexed]
DOI: 10.1007/978-3-031-04112-9_13
6. Aliev K., Narejo S., Pasero E. and Inoyatkhodjaev J. "A Predictive Model of Artificial Neural Network for Fuel Consumption in Engine Control System." In *Multidisciplinary Approaches to Neural Computing*, SIST vol. 69. Springer, Cham, 2018. [Scopus-indexed]
DOI: 10.1007/978-3-319-56904-8_21
7. Narejo S. and Pasero E. "An Application of Internet Traffic Prediction with Deep Neural Network." In *Multidisciplinary Approaches to Neural Computing*, SIST vol. 69. Springer, Cham, 2018. [Scopus-indexed]

CONFERENCE PUBLICATIONS

1. Chandio S.A., Narejo S. and Shaikh A.M. "Stellar AI Variability Predictor for Brightness Fluctuations and Galactic Effects." 2026 1st International Conference on Innovations in Information and Communication Technologies (IICT), pp. 1-3. IEEE, 2026.
DOI: 10.1109/IICT68374.2026.11468461
2. Soho R.M., Narejo S., Kolmykova A., Sheikh M.Z., Baro E.N. and Qureshi H. "Deep Learning-Based Surface Defect Detection for Railway Track Safety." 2026 Global Conference on Wireless and Optical Technologies (GCWOT), pp. 1-5. IEEE, 2026.
DOI: 10.1109/GCWOT69191.2026.11499540
3. Narejo S., Khan M.T., Shaikh M.Z., Dhirani L.L. and Chowdhery B.S. "Optimizing Route Efficiency in Formula One (F1) Vehicles Using Reinforcement Learning Algorithms." IEEE 4th International Conference on AI in Cybersecurity (ICAIC), Houston, TX, USA, 2025, pp. 1-8.
DOI: 10.1109/ICAIC63015.2025.10848611
4. Jatoi S., Abro B., Jiskani S., Baloch Y.A., Narejo S. and Asma K. "Learning Through Vision: Image Recognition in Early Education." 2025 International Conference on Emerging Technologies in Electronics, Computing, and Communication (ICETECC), Jamshoro, Pakistan, 2025, pp. 1-5.
DOI: 10.1109/ICETECC65365.2025.11070249
5. Soho R.M., Narejo S., Sheikh M.Z., Khalid M., Baro E.N. and Manolova A. "Automated Railway Surface Defect Detection and Classification Using Transfer Learning." 2025 28th International Symposium on Wireless Personal Multimedia Communications (WPMC), Sofia, Bulgaria, 2025, pp. 1-5.
DOI: 10.1109/WPMC67460.2025.11351231
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